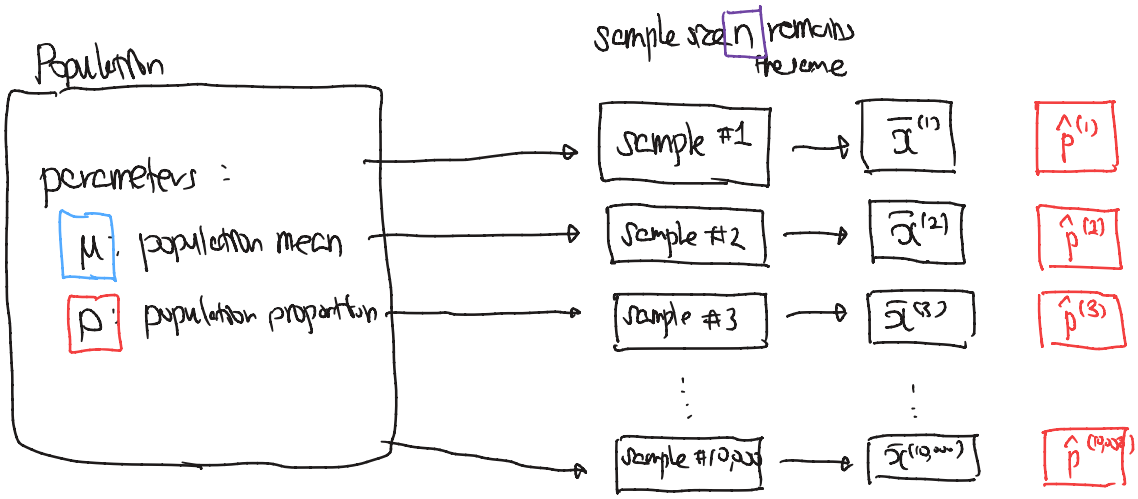
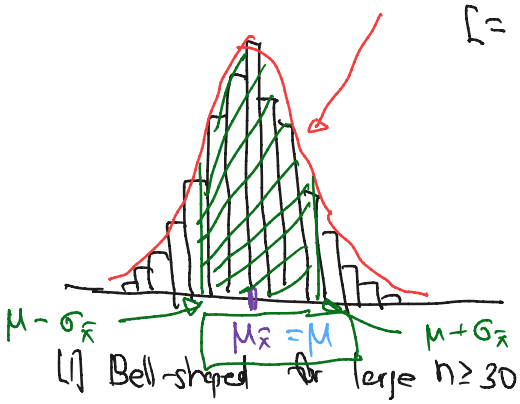
Sampling distributions



Ex: $\hat{p} = \frac{\#\text{ of people who have pets}}{\#\text{ of people chosen in a sample}} = \frac{20}{30}$

$\bar{x}^{(1)}, \bar{x}^{(2)}, \cdots, \bar{x}^{(10,000)}$

Histogram → Sampling distribution of $\bar{x}$ [= $\bar{x}$-distribution]



$\left[(\mu - \sigma_{\bar{x}}, \mu + \sigma_{\bar{x}}) \text{ contains middle } 68\% \text{ of } \bar{x}^{(1)}, \cdots, \bar{x}^{(10,000)} \right] \Rightarrow \left[\left(\mu - \frac{\sigma}{\sqrt{n}}, \mu + \frac{\sigma}{\sqrt{n}}\right) \text{ contains middle } 68\% \text{ of } \bar{x}^{(1)} \cdots \bar{x}^{(10,000)} \right]$

$\Downarrow$

$\left[\left(\mu - 2\cdot\frac{\sigma}{\sqrt{n}}, \mu + 2\cdot\frac{\sigma}{\sqrt{n}}\right) \text{ contains middle } 95\% \text{ of } \bar{x}^{(1)} \cdots \bar{x}^{(10,000)} \right]$

[1] Bell-shaped for large $n \geq 30$

[2] Center of the sampling dist. of $\bar{x}$

$\mu_{\bar{x}} = \mu$

↑ Avg. of sample means

95%

Note : $2 \cdot \frac{\sigma}{\sqrt{n}}$

$\frac{\sigma}{\sqrt{n}}$: standard error of $\bar{x}$-dist. (SE)

[3] Standard deviation of $\bar{x}$-distribution :

Standard error $= \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$ [σ denotes the standard deviation for the population]

↑ variability of $\bar{x}$'s from sample to sample

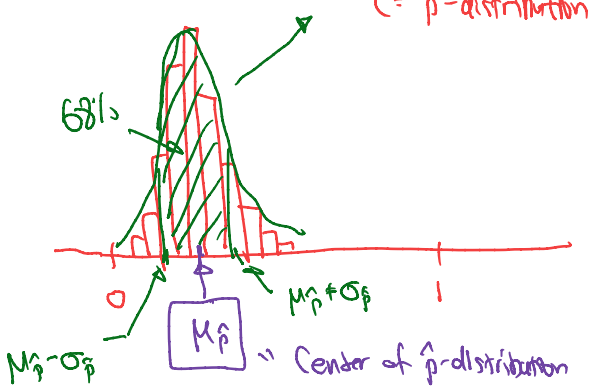
Note : $x_1, \cdots, x_n \rightarrow \bar{x} = \frac{\text{Sum of all } x_i\text{'s}}{\#\text{ of } \bar{x}\text{'s values}} = \frac{\Sigma x_i}{n}$

sample standard deviation → $s = \sqrt{\frac{\text{Sum of squared deviations}}{n-1}} = \sqrt{\frac{\Sigma (x_i - \bar{x})^2}{n-1}}$ ←

↑ variability of $x_1, \cdots, x_n$ in a single sample

$\hat{p}^{(1)}, \cdots, \hat{p}^{(10,000)}$

Histogram → sampling distribution of $\hat{p}$ (= $\hat{p}$-distribution)



$\left[(\mu_{\hat{p}} - \sigma_{\hat{p}},\ \mu_{\hat{p}} + \sigma_{\hat{p}}) \text{ contains middle 68\% of } \hat{p}^{(1)}, \cdots, \hat{p}^{(10,000)} \right] \Rightarrow$

68%

$\left[\left(p - \sqrt{\frac{p(1-p)}{n}},\ p + \sqrt{\frac{p(1-p)}{n}}\right) \text{ contains middle 68\% of } \hat{p}^{(1)}, \cdots, \hat{p}^{(10,000)} \right]$

$\left[\left(p - 2\sqrt{\frac{p(1-p)}{n}},\ p + 2\sqrt{\frac{p(1-p)}{n}}\right) \text{ contains middle 95\% of } \hat{p}^{(1)}, \cdots, \hat{p}^{(10,000)} \right]$

95%

$2\sqrt{\frac{p(1-p)}{n}}$

Standard error of $\hat{p}$-distribution

[1] Bell-shaped for large $n \geq 30$

[2] Center of $\hat{p}^{(1)}, \cdots, \hat{p}^{(10,000)}$ (= center of $\hat{p}$-distribution)

$\mu_{\hat{p}} = p$: population proportion

[3] Standard deviation of $\hat{p}^{(1)}, \cdots, \hat{p}^{(10,000)}$

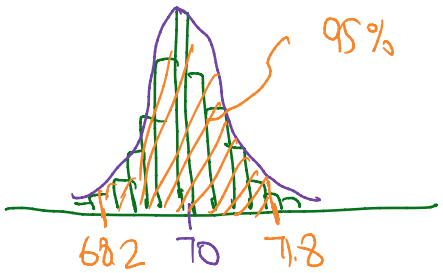
= Standard error of $\hat{p}$ = $\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$

Ex : μ = pop. mean = average height of people in a pop. = 70

σ = pop. standard deviation = standard deviation of heights in pop. = 9 in

$n = 100$

What is the sampling dist. of $\bar{x}$?



[1] Bell-shaped bdc $n \geq 30$

[2] Center of $\bar{x}$-dist. : $\mu_{\bar{x}} = \mu = 70$

[3] Standard error of $\bar{x}$: $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{9}{\sqrt{100}} = \frac{9}{10} = 0.9 = SE$

95% interval : $\mu \pm 2 \cdot SE$

$= 70 \pm 2 \cdot 0.9$

$= 70 \pm 1.8$

$= (68.2,\ 71.8)$ contains 95% of all sample means

Ex p = pop. proportion = proportion of people smoking at least 5 times per week = 0.4

$n = 100$

What is the sampling dist. of $\hat{p}$?

Note: $\hat{p} = \dfrac{\#\text{ of people smoking at least 5 times / week}}{\#\text{ of people in a sample}} = \dfrac{\square}{100}$

[1] Bell-shaped (normal) b/c $n \geq 30$

[2] Center of $\hat{p}$-distribution

$\mu_{\hat{p}} = p = 0.4$

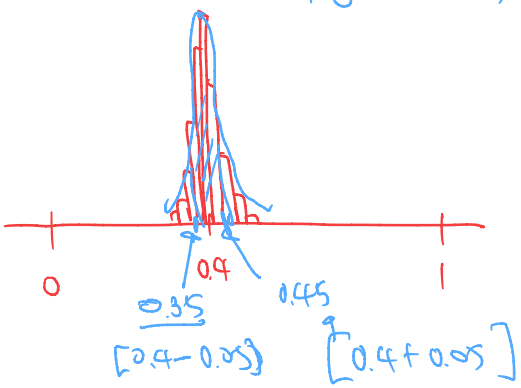
[3] Standard error of $\hat{p}$-distribution

$$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{(0.4)(1-0.4)}{100}} = \sqrt{\frac{0.4 \times 0.6}{100}} = \sqrt{\frac{0.24}{100}} = \sqrt{.0024} = 0.04898$$

$= 0.049$

≈ 0.05

SE of $\hat{p}$



(0.35, 0.45) contains 68% of middle $\hat{p}$ values

$p \pm 2 \cdot SE = 0.4 \pm 2(0.05)$

$= 0.4 \pm 0.10$

$= (.3, .5)$

(0.3, 0.5) covers middle 95% of $\hat{p}$ values.